

Keshav Rajput

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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Big Data, Principles of Database Systems, Information Visualization, Foundations of Data Science, Statistical Inference, Machine Learning

PROJECTS

NYC AutoDDG | Python, PySpark, pandas, PyArrow, Ollama

github.com/happyc0der/nyc-autoddg

- Single-pass PySpark profiler (types, statistics, semantic types, quality flags) for 100 NYC Open Data datasets: 4.8M rows in 2.3 min, down from 12.
- Benchmarked Spark vs pandas on an M4 Pro and Databricks serverless: crossover at 1.7M rows, near-linear to 8 cores, pandas 14x faster on small files.
- A local qwen3:14b wrote AutoDDG-style descriptions from the profile; 97% of checkable claims matched, and search Recall@1 rose from 0.605 to 0.728.

TrafficFlowBench (not yet public) | Python, LightGBM, scikit-learn, NumPy, SciPy

2026

- Kaggle TrafficFlowBench (IEEE Big Data Cup 2026): public leaderboard 0.845 vs the 0.553 baseline, up from 0.67365 on my first submission; final ranking pending.
- Same-link interpolation for missing cells, LightGBM queue forecasters trained out of core on 56M rows with causal features, exact-fit NNLS origin-destination solver.
- Found a window-selection regularity that lifted queue-onset IoU from 0.57 to 0.93; reported it on the forum, and the hosts announced it to all teams.

Kaggle S6E9, EV purchases | Python, LightGBM, XGBoost, CatBoost, PyTorch

github.com/happyc0der/kaggle-s6e9-ev-purchases

- Kaggle Playground S6E9 (ROC AUC): out-of-fold 0.94648, public leaderboard 0.94649 against a leader at 0.94674; final ranking pending.
- Nested target encodings over the synthetic generator's artifacts on 668,665 rows, 20 folds; logit blend of LightGBM, CatBoost and a 15-seed MLP.
- 192 experiments judged by paired DeLong tests; label resampling showed a perfect model scores 0.9459 ± 0.0003 , so the gap left is label noise.

Biohub cell tracking (in progress) | Python, PyTorch, NumPy, SciPy, polars

2026

The submission is a fork of Igor Zharov's public Harmonic Fusion notebook (Apache-2.0). The offline harness, the division classifiers and every experiment are mine.

- Kaggle Biohub Cell Tracking: public leaderboard 0.952 against 0.947 for the public notebook it builds on, on free Kaggle GPU; final ranking pending.
- Offline harness runs GPU inference once, then re-tunes post-processing locally against the official scorer on 40 held-out videos, matching the notebook to +0.00011.
- 3D CNN trained on 134 annotated divisions prunes false forks: predicted +0.0047 on a pre-registered holdout, measured +0.005 on the leaderboard twice.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- **Kaggle Playground S6E9**, 2026. Out-of-fold AUC 0.9465. Final ranking pending, competition ends 2026-09-30.
- **TrafficFlowBench**, 2026. Public leaderboard 0.845 against a 0.553 baseline. In progress, ends 2026-11-06.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

ML: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, vLLM, Ollama, NumPy, pandas, polars

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale

ACTIVITIES

- Events head, Finnexia (finance club), IIIT Delhi; organised "The Critical Bid"